Block reference

Calls, options, and actual output

Use this while writing a lecture. The gray area is code to copy; the bordered area shows the result produced by the current package. Click a topic below or open this PDF's bookmarks sidebar to jump to its reference page.

FIND THE RIGHT BLOCK

Task	Command or environment	Page
Organize topics	`\noteheading`; heading aliases	2
Navigate the PDF	`\notesetup`; contents and bookmarks	3
Explain or define	`\notebox`, `\notedefinition`, `\assumption`	4
Highlight a result	`\keyequation`	5
Explain a derivation	`notederivation`, `\derivestep`	6
Align compact steps	Legacy and multiline derivations	7
List values or symbols	`notetable`, `notation`	8
Place content side by side	`notepanels`, `notepanel`	9
Insert an existing image	`\notefigure`	10
Draw a circuit	`notecircuit`; presets and annotations	11
Plot a signal	`notesignal`	12
Include source code	`notecode`	13
Summarize a topic	`\notesummary` (School-Notes only)	14
Check or update lectures	`notes check`, `notes update-style`	15
Type blocks quickly	VS Code shortcuts and common mistakes	16

FIRST DOCUMENT

Place `lecturenotes.sty` beside `lecture.tex`. All block examples go between `\begin{document}` and `\end{document}`.

```
\documentclass[a4paper,11pt]{article}
\usepackage{lecturenotes}
\lectureheader{ELE734}{1}
\begin{document}
\tableofcontents
\clearpage
\noteheading{1}{First topic}
Your notes go here.
\end{document}
```

Reading the syntax. Braces `{...}` hold required arguments; brackets `[...]` hold optional arguments. An environment needs matching `\begin` and `\end`. In option lists, separate settings with commas and brace values that contain commas. Compile twice after changing headings or references. Existing lectures need the v0.2.0 style for the new blocks.

Headings

`\noteheading[options]{level}{title}`

Use levels 1 through 5 to express increasingly specific topics. The heading itself is unnumbered. Start with level 1 and move down one level at a time.

CALL

```
\noteheading{1}{Current mirrors}
\noteheading{2}{Cascode mirrors}
\noteheading[bookmark={Output resistance ro}]{3}
  {Output resistance $r_o$}
\noteheading{4}{Dominant term}
\noteheading{5}{Approximation condition}
```

RENDERED RESULT

Current mirrors

Cascode mirrors

Output resistance r_o

Dominant term

Approximation condition

OPTIONS AND DEFAULTS

`level`	Required integer from 1 to 5. Other levels cause an error.
`bookmark={...}`	Optional plain-text sidebar title. The printed heading keeps its mathematics; the sidebar gets a readable title.

EQUIVALENT OLDER NAMES

```
\notesection{Title}        % same as \noteheading{1}{Title}
\subnotesection{Title}     % same as \noteheading{2}{Title}
\subsubnotesection{Title}  % same as \noteheading{3}{Title}
```

Use `\noteheading{4}` and `\noteheading{5}` for deeper headings; there is no need to repeat “sub” in the command name.

Editor shortcuts: `;h1` through `;h5`; `;sec` and `;sub` for the older names. `;hbm` asks for the bookmark title, then the level, then the printed title.

Contents and PDF navigation

`\notesetup{toc-depth=...,bookmark-depth=...}`

Contents is a page printed inside your lecture PDF. **Bookmarks** are a topic tree in the PDF viewer's sidebar. Clicking either kind of entry jumps to the heading. These settings control the two lists independently; every heading still appears in the lecture body.

CALL: TWO LEVELS ON PAPER, FIVE IN THE SIDEBAR

```
% Set this before \tableofcontents and before your headings.
\notesetup{toc-depth=2,bookmark-depth=5}
\tableofcontents
\clearpage
% Now add the five headings shown on the previous page.
```

PRINTED CONTENTS: ONLY LEVELS 1 AND 2

RENDERED RESULT



SIDEBAR TREE FOR THE SAME FIVE HEADINGS

```
Current mirrors
  Cascode mirrors
    Output resistance ro
      Dominant term
        Approximation condition
```

OPTIONS AND DEFAULTS

`toc-depth=5`	Default. Values 0–5; 0 hides all note-heading entries from the printed contents. You still need `\tableofcontents` to print it.
`bookmark-depth=5`	Default. Values 0–5; 0 hides note-heading entries from the viewer's sidebar. It does not remove body headings.
`circuit-preset=none`	Another `\notesetup` setting. See page 11 for `compact` and `standard`.

When the list looks stale: compile twice. Open your PDF viewer's bookmarks or outline sidebar to see bookmarks. The optional heading `bookmark` text changes the sidebar title; the printed contents keeps the printed heading text. **Shortcut:** `;nav`.

Text, definitions, and assumptions

Three blocks for three kinds of prose

General explanation: `\notebox{title}{body}`. Use for a physical interpretation, a reminder, or a grouped explanation.

CALL

```
\notebox{Physical interpretation}{
  The capacitor stores charge. Rapid input changes are smoothed
  because its voltage cannot change instantaneously.
}
```

RENDERED RESULT

> **Physical interpretation**
> The capacitor stores charge. Rapid input changes are smoothed because its voltage cannot change instantaneously.

A term and its meaning: `\notedefinition{term}{body}`. The definition label and slim rule distinguish it from a general note.

CALL

```
\notedefinition{Time constant}{
  The quantity $\tau=RC$ sets the response speed of an RC circuit.
  After one time constant, a charging capacitor reaches about
  $63.2\%$ of its final voltage.
}
```

RENDERED RESULT

> DEFINITION **Time constant**
> The quantity $\tau = RC$ sets the response speed of an RC circuit. After one time constant, a charging capacitor reaches about 63.2% of its final voltage.

A condition on the model: `\assumption{body}`. This stays inline rather than taking the space of another box.

CALL

```
\assumption{Ideal components, no output load, and an initially
  uncharged capacitor.}
```

RENDERED RESULT

Assumption Ideal components, no output load, and an initially uncharged capacitor.

Bodies accept normal LaTeX prose, inline mathematics such as `$\tau =RC$`, and paragraphs. The boxes can continue onto another page. These three commands have no block-specific optional arguments.

Shortcut: `;def` inserts a definition. There are currently no dedicated block shortcuts for `\notebox` or `\assumption`.

Key equations

`\keyequation{title}{math}[explanation]`

Use for a result worth finding again. The equation is displayed in a tinted box, with an optional prose explanation below it.

CALL

```
\keyequation{RC low-pass transfer function}{
  H(s)=\frac{1}{1+sRC},\qquad f_c=\frac{1}{2\pi RC}
}[Take the output across the capacitor. Frequencies above
  $f_c$ are increasingly attenuated.]
```

RENDERED RESULT

RC low-pass transfer function

$$H(s) = \frac{1}{1 + sRC}, \qquad f_c = \frac{1}{2\pi RC}$$

Take the output across the capacitor. Frequencies above f_c are increasingly attenuated.

Argument order matters: the optional explanation comes *after* the title and equation. Omitting it gives the original two-argument call. The equation argument is already in math mode: do not surround it with `$...$`, `\[...\]`, or an `equation` environment.

MORE THAN ONE EQUATION LINE

Use `aligned` inside the equation argument. Put `&` where the lines should align, and `\\` at each line break.

CALL

```
\keyequation{Charging response}{
  \begin{aligned}
    v_C(t) &= V_0\left(1-e^{-t/\tau}\right) \\
    \tau   &= RC
  \end{aligned}
}
```

RENDERED RESULT

Charging response

$$\begin{aligned} v_C(t) &= V_0\left(1-e^{-t/\tau}\right) \\ \tau &= RC \end{aligned}$$

Numbering: key equations are unnumbered. For an equation that needs a numbered cross-reference, use a normal LaTeX `equation` environment and `\label{eq:unique-name}`.

Shortcut: `;keq` inserts the title and equation. Add the final `[explanation]` yourself when needed.

Derivations with explanations

`notederivation` containing `\derivestep{math}{prose}`

Use a step for each equation and the explanation that belongs to it. Explanations wrap as ordinary prose. Each step stays together; the full derivation can break between steps.

CALL

```
\begin{notederivation}[mode=steps]{Voltage gain}
  \derivestep{i_d=g_m v_{gs}}
    {Use the small-signal drain-current relation.}
  \derivestep{A_v=-g_m R_D}
    {The drain resistor converts current to voltage. The minus
     sign expresses the inversion between the input and output.}
\end{notederivation}
```

RENDERED RESULT

Voltage gain

$$i_d = g_m v_{gs}$$

Use the small-signal drain-current relation.

$$A_v = -g_m R_D$$

The drain resistor converts current to voltage. The minus sign expresses the inversion between the input and output.

PUT THE EXPLANATION BESIDE THE EQUATION

CALL

```
\begin{notederivation}[mode=steps,explanation=beside,
  equation-width=0.48\linewidth,gap=6mm]{Voltage gain}
  \derivestep{A_v=-g_m R_D}
    {The drain resistor converts current to voltage. This text
     wraps in the column beside the equation.}
\end{notederivation}
```

RENDERED RESULT

Voltage gain

$$A_v = -g_m R_D$$

The drain resistor converts current to voltage. This text wraps in the column beside the equation.

OPTIONS AND DEFAULTS

`mode=auto`	Default. Detects a literal `\derivestep` in the body. Use `mode=steps` when using `\input` or to state the layout explicitly.
`explanation=below`	Default. Choose `beside` for a text column on the right. `explanations` is an accepted alias.
`equation-width`	Default `0.52\linewidth`; left-column width in `beside` mode. Leave room for the explanation and the gap.
`gap=6mm`	Default. Space between columns in `beside` mode.

Shortcuts: `;drv` for the whole block; `;step` for another step. The first math argument is already in math mode; the second is prose.

Compact and multiline derivations

The original alignment remains supported

Compact explanations. Use the original `alignedat` body for short annotations that fit on one line. The initial `&` starts the equation column; `&&` starts its annotation column.

CALL

```
\begin{notederivation}[mode=legacy]{Cutoff frequency}
  & \omega_c=\frac{1}{RC}
    && \quad\text{radians per second} \\
  & f_c=\frac{\omega_c}{2\pi}
    && \quad\text{cycles per second}
\end{notederivation}
```

RENDERED RESULT

Cutoff frequency

$$\omega_c = \frac{1}{RC} \quad \text{radians per second}$$
$$f_c = \frac{\omega_c}{2\pi} \quad \text{cycles per second}$$

`mode=legacy` states the layout explicitly. Omitting the option also selects it when the body has no literal `\derivestep`. Annotations use `\text{...}` and do not wrap like prose. For longer explanations, use the step layout on page 6.

MULTIPLE MATH LINES IN ONE WRAPPING STEP

Inside `\derivestep`, `&` aligns equal signs and `\\` starts another math line. Do not add another `aligned` wrapper; the step already supplies one.

CALL

```
\begin{notederivation}[mode=steps]{Collect the terms}
  \derivestep{
    (\tau+T_s)y[n] &= T_s x[n]+\tau y[n-1] \\
    y[n] &= \alpha x[n]+(1-\alpha)y[n-1]
  }{Divide by $\tau+T_s$ and define $\alpha=T_s/(\tau+T_s)$.
    Both lines belong to one step and stay with this explanation.}
\end{notederivation}
```

RENDERED RESULT

Collect the terms

$$(\tau + T_s)y[n] = T_s x[n] + \tau y[n-1]$$
$$y[n] = \alpha x[n] + (1-\alpha)y[n-1]$$

Divide by $\tau + T_s$ and define $\alpha = T_s/(\tau + T_s)$. Both lines belong to one step and stay with this explanation.

Loading steps from a separate file:

```
\begin{notederivation}[mode=steps]{Voltage gain}
  \input{gain-steps.tex} % the file contains \derivestep calls
\end{notederivation}
```

Shortcut: `;der` inserts the original aligned-column layout. Use `;drv` for wrapping explanations.

Tables and notation

`notetable` for custom columns; `notation` for symbols

`\begin{notetable}{column specification}{header cells}` starts a full-width table. Use `l`, `c`, or `r` for natural-width columns; `X` makes a paragraph column that fills the remaining width.

CALL

```
\begin{notetable}{lXr}{
  \textbf{Quantity} & \textbf{Meaning} & \textbf{Value}}
  $R$ & Series resistance & $1\,\mathrm{k}\Omega$ \\
  $C$ & Shunt capacitance & $100\,\mathrm{nF}$ \\
  $\tau$ & Time constant & $100\,\mu\mathrm{s}$ \\
\end{notetable}
```

RENDERED RESULT

Quantity	Meaning	Value
R	Series resistance	$1\,\mathrm{k}\Omega$
C	Shunt capacitance	$100\,\mathrm{nF}$
τ	Time constant	$100\,\mu\mathrm{s}$

Separate cells with `&`. End each body row with `\\` or `\tabularnewline`. The header's row ending and the horizontal rules are automatic: do not add a trailing `\\` to the header argument. The number of cells must match the column specification.

A SYMBOL LIST WITH FIXED HEADERS

`notation` supplies three columns: Symbol, Meaning, and Unit. The middle column wraps. Write three cells per row.

CALL

```
\begin{notation}
  $T_s$ & Sampling period & $\mathrm{s}$ \\
  $f_s$ & Sampling frequency & $\mathrm{Hz}$ \\
  $\alpha$ & Discrete filter coefficient & dimensionless \\
\end{notation}
```

RENDERED RESULT

Symbol	Meaning	Unit
T_s	Sampling period	s
f_s	Sampling frequency	Hz
α	Discrete filter coefficient	dimensionless

Tables do not break across pages. Keep long lists in separate tables. For a literal ampersand in prose, write `\&` so it is not treated as a new cell.

Shortcut: `;tbl` inserts a custom notes table. There is no dedicated `notation` shortcut.

Content side by side

`notepanels` contains two or three `notepanel` columns

Put a diagram beside its explanation, compare two models, or show three stages. Panel titles are required. Widths are relative weights, not lengths.

CALL

```
\begin{notepanels}[columns=2,widths={2,1},gap=6mm]
  \begin{notepanel}{Model}
    A first-order low-pass filter has one time constant.
    Its output follows slow changes and smooths rapid changes.
    \[H(s)=\frac{1}{1+sRC}\]
  \end{notepanel}
  \begin{notepanel}{Interpretation}
    Larger $RC$ means a slower response and a lower cutoff.
  \end{notepanel}
\end{notepanels}
```

RENDERED RESULT

Model

A first-order low-pass filter has one time constant. Its output follows slow changes and smooths rapid changes.

$$H(s) = \frac{1}{1 + sRC}$$

Interpretation

Larger RC means a slower response and a lower cutoff.

THREE EQUAL COLUMNS

CALL

```
\begin{notepanels}[columns=3]
  \begin{notepanel}{Input} $x[n]$: new sample. \end{notepanel}
  \begin{notepanel}{State} $y[n-1]$: previous output. \end{notepanel}
  \begin{notepanel}{Output} $y[n]$: updated output. \end{notepanel}
\end{notepanels}
```

RENDERED RESULT

Input

$x[n]$: new sample.

State

$y[n-1]$: previous output.

Output

$y[n]$: updated output.

OPTIONS AND DEFAULTS

`columns=2`	Default. Only 2 or 3 is supported. Additional panels start another row after this many columns.
`widths={}`	Default: equal widths. `widths={2,1}` gives the first panel twice the usable width. Supply one positive number per column.
`gap=6mm`	Default. Gaps are removed before widths are distributed.
`align=top`	Default. Also `center` or `bottom`.
`keep-together=true`	Default. Keeps the entire block on one page. `false` permits breaks between complete rows.

Each panel is a minipage and cannot split internally. Use inline figures (`placement=here`) inside it; normal floats cannot go inside a minipage. A panel must be inside `notepanels`. Leave enough space for the whole row. **Shortcuts:** `;pan2`, `;pan3`, and `;panel`.

Images with captions

`\notefigure[options]{image file}{caption}`

Use for an existing PNG, JPEG, or PDF figure. It fits within both dimension limits, preserves its aspect ratio, and adds a numbered caption.

CALL

```
\notefigure[width=0.8\linewidth,
  placement=here,label=fig:rc-response]
  {docs/reference/figures/rc-response.pdf}
  {Normalized response to a unit step.}

The curve in Figure~\ref{fig:rc-response} approaches one.
```

RENDERED RESULT

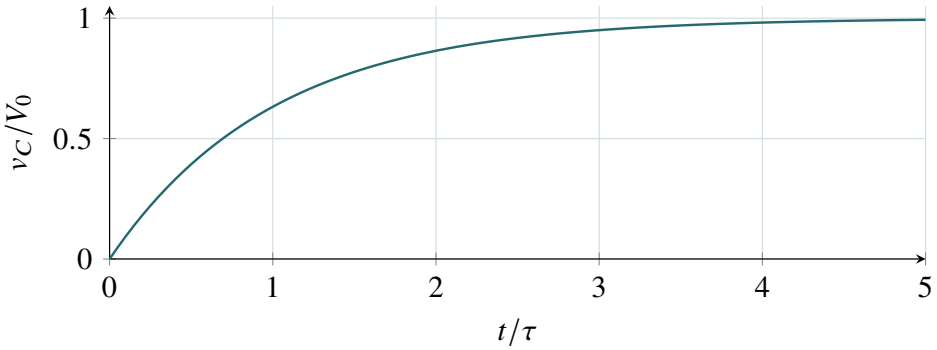


Figure 1: Normalized response to a unit step.

The curve in Figure 1 approaches one.

OPTIONS AND DEFAULTS

`width=0.85\linewidth`	Default maximum width. `\linewidth` adapts to the current panel or text area.
`maxheight=65mm`	Default maximum height. The smaller of the two bounds determines the final scale; the image is not stretched.
`placement=here`	Default. Image and caption stay together, inline at the call. Use this inside panels.
`placement=float`	Uses normal `htbp` floating. Placement letters such as `t` or `htbp!` are also supported; LaTeX chooses the final position.
`label={}`	Default: no reference target. Set a unique name such as `fig:rc-response` and use `\ref{fig:rc-response}`.

The example's long path is for this reference's included asset. In a lecture, put the image in `figures/` and use `{figures/rc-response.pdf}`. Paths are relative to the document's build directory. You may set `\graphicspath{{figures/}}` and use just the file name. Both dimensions must be positive. Compile twice to resolve figure numbers.

Shortcut: `;fig` asks for the file name, caption, and label suffix. It supplies `figures/`, `fig:`, width `0.8\linewidth`, and inline placement. Replace every field with your actual values.

Circuits and annotations

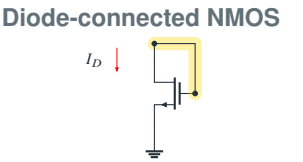
`\begin{notecircuit}[drawing options]{title}`

Write ordinary CircuitikZ commands inside the block. Set a preset before the diagram to avoid repeating the common scale and symbol options.

CALL

```
\notesetup{circuit-preset=compact}
\begin{notecircuit}{Diode-connected NMOS}
  \node[nmos,arrowmos,xscale=-1] (M) at (0,0) {};
  \coordinate (bias) at (0,1.2);
  \draw[note-highlight]
    (bias)--(M.G |- bias)--(M.G);
  \draw (M.D)--(bias) node[circ]{}
    --(M.G |- bias)--(M.G) node[circ]{};
  \draw (M.S)--++(0,-0.2) node[ground]{};
  \draw[note-current] (-0.9,1.1)--(-0.9,0.5);
  \node[left] at (-1.1,0.8) {$I_D$};
\end{notecircuit}
```

RENDERED RESULT



OPTIONS AND DEFAULTS

`circuit-preset=none`	Default. Supplies no drawing options; existing per-diagram settings remain in control.
`circuit-preset=compact`	American symbols, scale 0.55, transformed labels, and compact current arrows. A useful starting point for notes.
`circuit-preset=standard`	Same conventions at scale 0.8.
`note-highlight`	TikZ style: translucent yellow, rounded, 5pt stroke. Draw it before the dark wire so the connection remains clear.
`note-current`	TikZ style: thin red current arrow with a small tip. Put its label in a separate node with a little space from the arrow.

Orthogonal MOS connection. `(M.G |- bias)` uses the gate's horizontal coordinate and the bias node's vertical coordinate. Connecting through that point gives horizontal and vertical wires even when the symbol is mirrored. It avoids a guessed gate coordinate and an angled wire.

Explicit drawing options override the preset, for example `\begin{notecircuit}[scale=0.65]{Title}`. The block centers the diagram and adds its title; it does not number a figure or create a floating caption. Keep wire runs short and give labels space. **Shortcuts:** `;cset`, `;cir`, `;cur`, `;mark`.

Signal and response plots

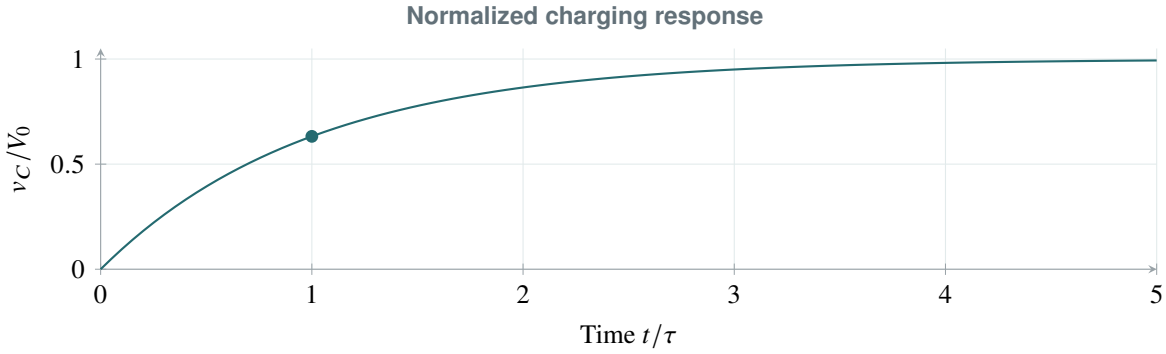
`\begin{notesignal}[axis options]{title}`

Use for a mathematical curve or a set of data points. The block provides a centered PGFPlots axis, muted grid, and teal plot strokes.

CALL

```
\begin{notesignal}[
  xlabel={Time $t/\tau$},ylabel={$v_C/V_0$},
  xmin=0,xmax=5,ymin=0,ymax=1.05,
  xtick={0,1,2,3,4,5},ytick={0,0.5,1},height=45mm]
  {Normalized charging response}
  \addplot[domain=0:5,samples=100] {1-exp(-x)};
  \addplot[only marks,mark=*]
    coordinates {(1,0.63212)};
\end{notesignal}
```

RENDERED RESULT



OPTIONS YOU WILL USUALLY CHANGE

OPTIONS AND DEFAULTS

`xlabel`, `ylabel`	Axis labels. Brace values containing math, spaces, or commas, as shown above.
`xmin`, `xmax`	Visible horizontal range. Use `ymin` and `ymax` for the vertical range.
`xtick`, `ytick`	Explicit tick lists, for example `xtick={0,1,2}`.
`width`, `height`	Defaults `0.94\linewidth` and `5cm`. The example uses a slightly shorter axis.
`domain`, `samples`	Options on `\addplot` for a formula, not required arguments of the block.

The optional list accepts normal PGFPlots axis settings. Use PGFPlots expression syntax inside a formula: `1-exp(-x)`, not a LaTeX equation. For data, use `\addplot coordinates {(0,0) (1,0.5) (2,0.8)};`. Add another `\addplot` for another curve and `\addlegendentry{...}` after a plot for a legend entry. End drawing commands with semicolons. There is no dedicated `notesignal` shortcut.

Source-code blocks

`\begin{notecode}[listings options]`

Code is read verbatim: type underscores, braces, and # normally. The block supplies a neutral background, thin frame, syntax colors, and automatic line wrapping.

CALL

```
\begin{notecode}[language=C,numbers=left]
float lowpass_step(float x, float previous_y, float alpha)
{
    return previous_y + alpha * (x - previous_y);
}
\end{notecode}
```

RENDERED RESULT

```c
float lowpass_step(float x, float previous_y, float alpha)
{
    return previous_y + alpha * (x - previous_y);
}
```

ARM ASSEMBLY

CALL

```
\begin{notecode}[language=ARM]
MOVS r0, #1       ; Start with one bit
LSLS r0, r0, #5   ; Move it to bit 5
ORRS r1, r1, r0   ; Set that bit
\end{notecode}
```

RENDERED RESULT

```
MOVS r0, #1       ; Start with one bit
LSLS r0, r0, #5   ; Move it to bit 5
ORRS r1, r1, r0   ; Set that bit
```

OPTIONS AND DEFAULTS

`language=C`	Default. Choose another language supported by `listings`, or the bundled `ARM` lexer.
`numbers=none`	Default. `numbers=left` adds line numbers.
`firstnumber=1`	Standard listings option; change the starting line number when presenting a later part of a program.

The ARM lexer highlights a small core of instructions and `;` comments; it is a display helper, not an assembler or validator. Place `notecode` directly in the document. Verbatim environments do not work inside a macro argument such as the body of `\notebox`, or inside the captured body of `notepanels`. Keep the closing `\end{notecode}` on its own line. There is no dedicated block shortcut.

School-Notes summary block

`\notesummary{title}{overview}{table rows}` - local extension

This command exists in the updated lecture styles in your School-Notes directory. It is preserved when updating those styles, but is **not** bundled into the style generated by `notes new`.

CALL

```
\notesummary{RC low-pass filter}{
  Take the output across the capacitor. The circuit passes
  slow changes and attenuates rapid changes.
}{
  Model & $H(s)=1/(1+sRC)$ \\
  Cutoff & $f_c=1/(2\pi RC)$ \\
  Limit & Assumes no load at the output. \\
}
```

RENDERED RESULT

> **RC low-pass filter**
> Take the output across the capacitor. The circuit passes slow changes and attenuates rapid changes.

Model	$H(s) = 1/(1 + sRC)$
Cutoff	$f_c = 1/(2\pi RC)$
Limit	Assumes no load at the output.

WHAT EACH ARGUMENT DOES

OPTIONS AND DEFAULTS

`title`	Required title at the top of the white note box.
`overview`	Required prose in that box. Inline math and paragraphs work like normal LaTeX.
`table rows`	Required two-column rows below the box. Use a short label, `&`, the description, then `\\`.

The label column is fixed at `4em`; the description column fills the remaining width and wraps. Keep labels short, such as Model, Limit, or Use. There are no automatically inserted table headings and no optional arguments.

IF THE COMMAND IS UNDEFINED IN A NEW LECTURE

Use `\notebox` followed by `notetable` for the same general structure, or intentionally copy the local definition into that lecture's style. The reference loads a separate copy of this extension only to show its output; the canonical package remains the source for every other example.

Editor note: `;sum` inserts a mathematical summation, not `\notesummary`. There is no summary-block shortcut at present.

Checking and updating lectures

Terminal commands that support the writing workflow

CHECK A LECTURE, COURSE, OR THE WHOLE NOTES ROOT

```
notes check ELE734/lecture-01
notes check ELE734 --compile
notes check . --strict
```

Run commands from your School-Notes root, or supply a path to it. Omitting the path checks the current directory. The static checker follows literal `\input` and `\include` files and reports duplicate labels, undefined references, missing files, placeholder labels, and unfinished notes.

ILLUSTRATIVE OUTPUT

```
ELE734/lecture-01/lecture.tex:42: error [undefined-reference]:
  reference "fig:gain" has no label
```

OPTIONS AND DEFAULTS

`-compile`	Runs pdfLaTeX twice without shell escape, in a temporary output directory. Reports build errors and layout warnings while preserving existing PDFs, logs, and auxiliary files. Each pass has a 45-second timeout.
`-strict`	Warnings also produce exit status 1. Normally warnings return 0, errors return 1, and invalid command syntax returns 2.

The static scan does not expand arbitrary TeX macros or know every package-generated label. Use the real build to assess those cases.

PREVIEW, THEN APPLY A STYLE UPDATE

```
notes update-style ELE734 --dry-run
notes update-style ELE734
```

This accepts a lecture, course, or notes root; omission means the current directory. The preview reports versions, custom edits, and conflicts. Applying merges compatible local edits into the current bundled style, including local colors and added commands such as `\notesummary`.

Concrete file change: `lecturenotes.sty` becomes the updated style. Its exact previous bytes go into `lecturenotes.sty.bak`, then `.bak.1`, `.bak.2`, and so on if earlier backups exist. Your `lecture.tex` and figures are unchanged. Already-current styles get no new backup. Rebuild the lecture afterward to see the updated formatting.

Conflicts: all lectures are checked before any style is replaced. Inspect the reported conflict first. `-resolve=keep` favors conflicting local edits; `-resolve=replace` favors the updated definition. Both retain compatible additions. Keeping an old definition can retain old behavior. Unknown versions and complex or structurally invalid merges need a manual merge.

Editor shortcuts and quick fixes

Keep this page open beside VS Code

In your School-Notes workspace, type these at the start of a line outside math mode. HyperSnips expands the block; use Tab to move through its fields. After updating snippets, run **HyperSnips: Reload Snippets** from the Command Palette. These shortcuts belong to the School-Notes workspace.

Shortcut	Inserts
`;h1–;h5`	Heading at the chosen level.
`;sec`, `;sub`	Main and second-level heading aliases.
`;hbm`	Heading with plain bookmark title; fields: bookmark, level, title.
`;nav`	Printed-contents depth and sidebar depth.
`;def`	Definition with a term and body.
`;keq`	Key equation with title and math.
`;der`	Original derivation with short aligned annotations.
`;drv`, `;step`	Wrapping derivation and an additional step.
`;tbl`	Table specification, header, and body row.
`;pan2`, `;pan3`	Two or three titled panels.
`;panel`	Additional panel inside `notepanels`.
`;fig`	Image, caption, and label; fields: file name, caption, label suffix.
`;cset`, `;cir`	Circuit preset and circuit block.
`;cur`, `;mark`	Current arrow and highlighted connection.

COMMON MISTAKES

Symptom	Fix
Undefined new command	Update that lecture's local style to v0.2.0. For `\notesummary`, see page 14.
An equation gives a math error	Remove the outer `$...$` or `\[...\]` from `\keyequation` or `\derivestep`.
Extra alignment tab error	Match each table row's number of cells to the columns. Escape literal ampersands as `\&`.
Figure number is "??"	Give the figure a unique label, use exactly the same name in `\ref`, and compile twice.
A panel or step runs off the page	Shorten or split its content. A panel or individual step cannot split internally.
Floating figure fails in a panel	Set `placement=here` for that figure.
Derivation loaded by `\input` fails	Set `mode=steps` on `notederivation`.

Copyable source: the illustrated block examples live in `docs/reference/examples/`. The reference includes those same files to render the results, so the calls and outputs stay in sync.